

WE FOCUS ON TWO ONE-SIDED BEHAVIORAL TRANSITIONS

Taxonomy turns aggregate model differences into semantically targeted cohorts

DEEPSEEK · 215 IMPROVEMENTS

base fail → fine-tuned pass

119/215: generated test/import contamination
48: wrong output or logic
16: missing name/import
32: other primary categories

Mechanistic focus: 80 contamination cases

CODELLAMA · 291 REGRESSIONS

base pass → merged fail

120/291: API/type mismatch
50: wrong logic/other runtime
43: edge case/exception
78: generation, import, syntax, commentary

Mechanistic focus: 50 wrong-logic/runtime cases

HOW WE CLASSIFIED

Rule/AI-assisted taxonomy + human spot checks

IMPORTANT QUALIFICATION

Primary = most salient observed error—not a unique root cause. CodeLlama failures are heterogeneous and oft

DEEPSEEK /5 · CONTAMINATION

BASE · FAIL

Correct-looking body continues into:

```
`#tests/test_task_func.py`  
`import pytest`
```

FINE-TUNED · PASS

Completes the function and stops after a short example.

First textual divergence $\approx 76\%$ of shorter code;
target boundary is late/post-solution.

CODELLAMA /490 · WRONG LOGIC/API CONTRACT

BASE · PASS

parses XML $\rightarrow$ writes JSON file $\rightarrow$ returns result

MERGED · FAIL

`return xmltodict.parse(s)`
omits the required file write side effect

First divergence $\approx 81\%$ of base code;
target region is the late function body.

EIGHT PRE-DECLARED TOP-10 LISTS DEFINE THE $\alpha=3$ CANDIDATE POOL

Association vs paired model $\Delta \times$ global vs error-local scope

DEEPSEEK · IMPROVEMENT

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
3602	900	11172	9009
10168	14653	1767	5456
5422	2021	16044	309
1982	7279	10532	12591
1862	16331	9632	9982
1650	2582	7027	6927
5039	12591	1395	1126
9580	2380	6565	900
5837	10257	14818	13220
9537	15715	6911	6356

40 nominations → 38 unique features

2 feature(s) appear in more than one list

CODELLAMA · REGRESSION

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
13147	4837	8138	3612
8992	2861	3612	8309
1840	12780	12253	5642
16161	4313	1840	6214
15471	10879	11227	13428
10570	10111	14765	1231
8138	8558	4148	13934
7353	71	2286	8965
14359	9579	12573	11222
2825	8545	12643	354

40 nominations → 37 unique features

3 feature(s) appear in more than one list

NEW $\alpha=3$ SWEEP: exactly 38 DeepSeek + 37 CodeLlama features · no post-hoc additions

Hard gate: baseline must reproduce the original raw completion byte for byte before any steered arm runs.